

## Alternating series test - 5.5

Definition: If  $(a_n)$  has terms alternating between positive and negative, then  $\sum_{n=1}^{\infty} a_n$  is called an alternating series.

Examples:  $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \dots$

$$\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n+3} = -\frac{1}{4} + \frac{4}{5} - \frac{9}{6} + \frac{16}{7} - \dots$$

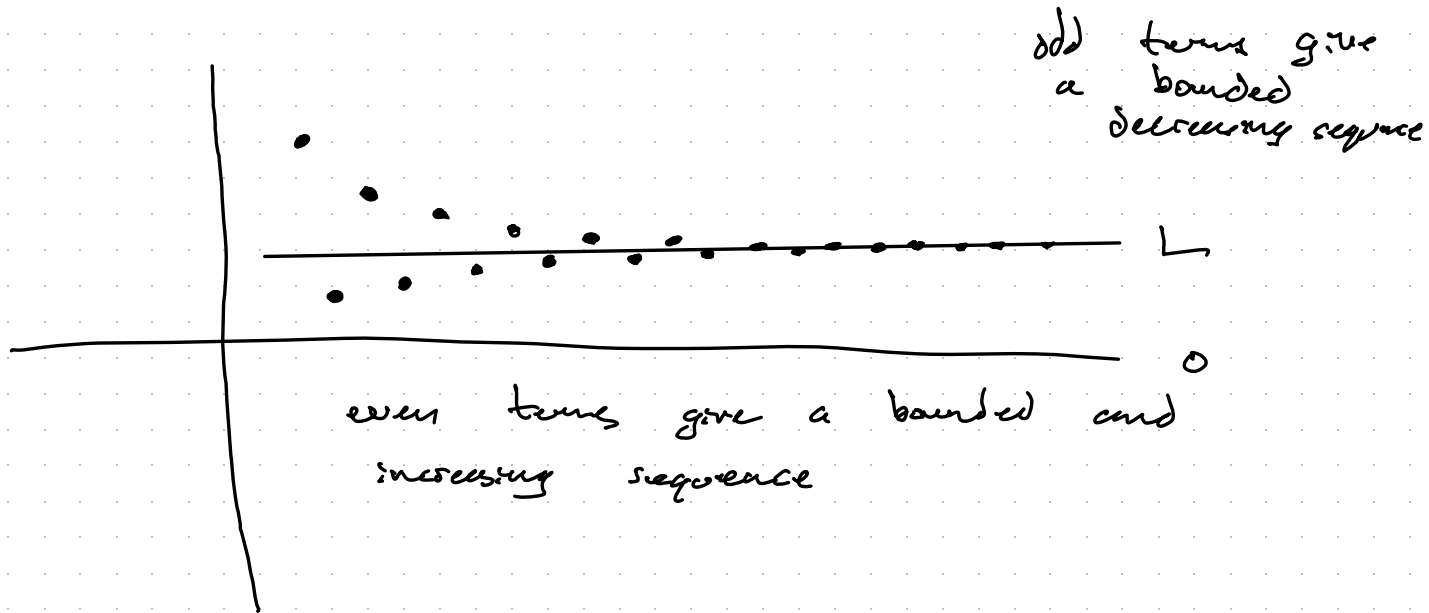
Every alternating series can be written

as  $\sum_{n=1}^{\infty} (-1)^n |a_n|$  or  $\sum_{n=1}^{\infty} (-1)^{n-1} |a_n|$

Alternating series test: If  $\sum_{n=1}^{\infty} a_n$  is an

alternating series with  $0 < |a_{n+1}| \leq |a_n|$  and

$\lim_{n \rightarrow \infty} |a_n| = 0$ , then it converges.



sketch of the sequence  $(s_k)$  of partial sums

Example :  $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{n^4+2}$  is alternating,

and  $|a_n| = \frac{n^3}{n^4+2}$  is decreasing to 0, so

by the alternating series test,  $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{n^4+2}$

converges.